



**ABES Engineering College, Ghaziabad**  
**Department of Applied Sciences & Humanities**

**Session: 2025-26**

**Semester: I**

**Course Code: 25AS102**

**Course Name: Fundamentals of Linear Algebra & Statistics**

**QUESTION BANK**

**UNIT-2**

**LINEAR TRANSFORMATION**

| S.No. | KL | CO  | Question                                                                                                                                                                                   | Marks |
|-------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1     | K3 | CO2 | Show that the function linear transformation $T: R^2 \rightarrow R^3$ is given by $T(x, y) = (x + y, x - y, y)$ is a linear transformation.                                                |       |
| 2.    | K3 | CO2 | Show that the function linear transformation $T: R^3 \rightarrow R^2$ is given by $T(x, y, z) = (2x - 3y, 7x + 2z)$ is a linear transformation                                             |       |
| 3.    | K3 | CO2 | Show that the function linear transformation $T: R^3 \rightarrow R^2$ is given by $T(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z)$ is a linear transformation.                                   |       |
| 4     | K3 | CO2 | Let V be vector space of real n-square matrices, and let M be a fixed non zero matrix in V. Show that the following mapping $T: V \rightarrow V$ is linear, defined by $T(A) = MA$         |       |
| 5     | K3 | CO2 | Show that the following mappings are not linear<br>(a) $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (x^2, y^2)$<br>(b) $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x + 1, y + z)$ |       |
| 6     | K3 | CO2 | Show that the following mappings are not linear<br>(a) $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (xy, y)$<br>(b) $T: R^2 \rightarrow R^2$ defined by $T(x, y) = ( x , y + z)$         |       |

|    |    |     |                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
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| 7  | K3 | CO2 | <p>Let <math>T: R^3 \rightarrow R^3</math> be the linear transformation defined by</p> $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$ <p>Find a basis and dimension of (a) Image of T (b) the kernel of T. Also verify rank nullity theorem.</p>                                                                                                                                                                   |  |
| 8  | K3 | CO2 | <p>Let <math>T: R^4 \rightarrow R^3</math> be the linear transformation defined by</p> $T(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t)$ <p>Find a basis and dimension of (a) Image of T (b) the kernel of T. Also verify rank nullity theorem.</p>                                                                                                                                                   |  |
| 9  | K3 | CO2 | <p>Consider the mapping <math>T: R^4 \rightarrow R^3</math>, where</p> $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix}$ <p>Find a basis and the dimension of</p> <p>(a) the image of A (b) the kernel of A.</p>                                                                                                                                                              |  |
| 10 |    | CO2 | <p>Consider the mapping <math>T: R^4 \rightarrow R^3</math>, where</p> $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -3 & 2 & -2 \end{bmatrix}$ <p>Find a basis and the dimension of</p> <p>(a) the image of A (b) the kernel of A.</p>                                                                                                                                                             |  |
| 11 |    | CO2 | <p>Consider the linear mapping <math>T: R^2 \rightarrow R^2</math> defined by</p> $T(x, y) = (3x + 4y, 2x - 5y)$ <p>and the following bases of <math>R^2</math>:</p> $S = \{u_1, u_2\} = \{(1, 2), (2, 3)\}$ $E = \{e_1, e_2\} = \{(1, 0), (0, 1)\}$ <p>(i) Find the matrix A representing T relative to basis S</p> <p>(ii) Find the matrix B representing T relative to basis E</p>                              |  |
| 12 |    | CO2 | <p>Consider the following operator <math>T</math> on <math>R^2</math> and basis S:</p> $T(x, y) = (2x - 7y, 4x + 5y)$ <p>and <math>S = \{u_1, u_2\} = \{(1, 3), (2, 5)\}</math></p> <p>(i) Find the matrix representation <math>[T]_S</math> of T relative to basis S.</p> <p>(ii) Verify <math>[T]_S [v]_S = [T(v)]_S</math> for the vector <math>v = (4, -3)</math> in <math>R^2</math>.</p>                     |  |
| 13 |    | CO2 | <p>Consider the following matrix A and basis of <math>R^2</math>:</p> $A = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix}$ <p>and <math>S = \{u_1, u_2\} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}</math></p> <p>The matrix A defines a linear operator on <math>R^2</math>.</p> <p>Find the matrix B that represents the mapping A T relative to basis S.</p> |  |
| 14 |    | CO2 | <p>Let V be the vector space of functions and let <math>D: V \rightarrow V</math> be the differential operator defined by <math>D\{f(t)\} = \frac{d}{dt}\{f(t)\}</math></p> <p>Find the matrix representing D in each basis:</p> <p>(i) <math>S = \{e^{3t}, te^{3t}, t^2e^{3t}\}</math> (ii) <math>S = \{\sin t, \cos t, e^{3t}\}</math></p>                                                                       |  |

|    |  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|----|--|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 15 |  | CO2 | <p>Let <math>T: R^3 \rightarrow R^2</math> be the linear transformation defined by<br/> <math>T(x, y, z) = (2x + y - z, 3x - 2y + 4z)</math><br/> Find the matrix of T in the following bases of <math>R^3</math> and <math>R^2</math> :<br/> <math>S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}</math> and<br/> <math>S' = \{u_1, u_2\} = \{(1, 3), (1, 4)\}</math></p>                                                                                                                                                                                                                                      |  |
| 16 |  | CO2 | <p>Let <math>A = \begin{bmatrix} 2 &amp; 5 &amp; -3 \\ 1 &amp; -4 &amp; 7 \end{bmatrix}</math>. Matrix A determines a mapping<br/> <math>T: R^3 \rightarrow R^2</math> defined by <math>A(v) = Av</math>, where vectors are written<br/> as columns.<br/> Find the matrix <math>[T]</math> that represents the mapping relative to<br/> the following bases of <math>R^3</math> and <math>R^2</math> :<br/> (i) <math>S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}</math> and<br/> <math>S' = \{u_1, u_2\} = \{(1, 3), (2, 5)\}</math><br/> (ii) The usual bases of <math>R^3</math> and <math>R^2</math></p> |  |
| 17 |  | CO2 | <p>Consider the following two bases of <math>R^2</math> :<br/> <math>S = \{u_1, u_2\} = \{(1, -2), (3, -4)\}</math><br/> and <math>S' = \{v_1, v_2\} = \{(1, 3), (3, 8)\}</math><br/> (i) Find the change-of-basis matrix P from S to <math>S'</math> .<br/> (ii) Find the change-of-basis matrix Q from <math>S'</math> back to S.<br/> (iii) Verify <math>Q = P^{-1}</math></p>                                                                                                                                                                                                                                            |  |
| 18 |  | CO2 | <p>The vectors <math>u_1 = (1, 2, 0)</math>, <math>u_2 = (1, 3, 2)</math>, <math>u_3 = (0, 1, 3)</math> form<br/> a basis S of <math>R^3</math>, Find<br/> (i) The change-of-basis matrix P from the usual basis<br/> <math>E = \{e_1, e_2, e_3\}</math> to S<br/> (ii) The change-of-basis matrix Q from S back to E.</p>                                                                                                                                                                                                                                                                                                   |  |
| 19 |  | CO2 | <p>(i) Let T be a linear operator on <math>R^2</math> defined by<br/> <math>T(x, y) = (2x - 3y, 3x - 4y)</math><br/> (a) Show that T is invertible<br/> (b) Find a formula for <math>T^{-1}</math>.<br/> (ii) Let <math>G: R^2 \rightarrow R^3</math> be defined by<br/> <math>G(x, y) = (x + y, x - 2y, 3x + y)</math><br/> (a) Show that G is non singular<br/> (b) Find a formula for <math>G^{-1}</math>.</p>                                                                                                                                                                                                            |  |
| 20 |  | CO2 | <p>Consider the linear operator on T on <math>R^3</math> defined by<br/> <math>T(x, y, z) = (2x, 4x - y, 2x + 3y - z)</math><br/> (a) Show that T is invertible<br/> (b) Find a formula for <math>T^{-1}</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |  |

Ans:

7. (a)  $(1, 0, 1), (0, -1, -1)$   $\dim(\text{Im } T) = 2$  (b)  $(3, -1, -1)$   $\dim(\text{Ker } T) = 1$   
8. (a)  $(1, 1, 1), (0, 1, 2)$   $\dim(\text{Im } T) = 2$  (b)  $(2, 1, -1, 0), (1, 2, 0, 1), (0, 1, 1, 0)$   $\dim(\text{Ker } T) = 2$   
9. (a)  $(1, 1, 3), (0, 1, 2)$   $\dim(\text{Im } A) = 2$ , (b)  $(1, -2, 1, 0), (-7, 3, 0, 1)$ ,  $\dim(\text{Ker } T) = 2$

10. (a)  $(1, 2, 1), (0, 1, 1)$   $\dim(\text{Im}A) = 2$ , (b)  $(4, -2, -5, 0), (1, -3, 0, 5)$ ,  $\dim(\text{Ker } T) = 2$

11. (i)  $[A]_s = \begin{bmatrix} -49 & -76 \\ 30 & 47 \end{bmatrix}$  (ii)  $[B]_E = \begin{bmatrix} 3 & 4 \\ 2 & -5 \end{bmatrix}$

12. (i)  $[T]_s = \begin{bmatrix} 121 & 201 \\ -70 & -116 \end{bmatrix}$

13.  $[B]_s = \begin{bmatrix} 7 & 14 \\ -4 & -9 \end{bmatrix}$

14. (i)  $[D]_s = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$  (ii)  $[D]_s = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

15.  $[T]_{s,s'} = \begin{bmatrix} 3 & 11 & 5 \\ -1 & -8 & -3 \end{bmatrix}$

16. (i)  $\begin{bmatrix} -12 & -41 & -8 \\ 8 & 24 & 5 \end{bmatrix}$ . (ii)  $\begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$ .

17. (i)  $P = \begin{bmatrix} -\frac{13}{2} & -18 \\ \frac{5}{2} & 7 \end{bmatrix}$  (ii)  $Q = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix}$

18. (i)  $P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$  (ii)  $Q = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}$

19. (i)  $T^{-1}(x, y) = (-4x + 3y, -3x + 4y)$  (ii)  $G^{-1}$  does not exist.

20.  $T^{-1}(x, y, z) = (\frac{1}{2}x, 2x - y, 7x - 3y - z)$